EOAT Command Center Feature & Implementation Expansion Plan

Purpose

This document defines the next major feature expansion plan for the EOAT Command Center repository.

The goal is not to add random buttons, random dashboards, or another glorious rectangle that says “Coming Soon” while contributing nothing to civilization. The goal is to turn the existing EOAT Command Center into a more guided, faster, more truthful, and more usable engineering workflow system for EOAT auditing, press/machine review, preventive maintenance tracking, compatibility management, documentation control, and final project handoff.

This document should be used by Codex as a feature planning and implementation guide.

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1. Current App Direction

Current Product Identity

The app is a local-first PySide desktop dashboard for EOAT project workflows. It supports:

- EOAT audit entry
- Workbook validation
- Audit progress tracking
- Press View
- Compatibility entries
- Notes and tags
- Open items
- Photo intake
- Issue analysis
- FMEA-lite
- KPI summaries
- Pilot candidate scoring
- PM checklist generation
- BOM / spare parts analysis
- Standards documentation review
- Scheduled daily/weekly summaries
- Final handoff packaging
- Backup management
- Release readiness checks
- Performance logging
- Dashboard caching

The repo should continue to keep real operational data outside the repository. The repo should only contain app source code, generic documentation, sanitized templates, tests, and synthetic demo data.

Real audit workbooks, machine data, capacity files, mold numbers, part numbers, maintenance histories, downtime/scrap/cycle-time exports, customer names, generated reports, and internal shared-drive paths must remain outside the repo.

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2. Product-Level Problem Statement

The current app has become powerful, but it is also becoming large enough that usability and workflow clarity now matter more than simply adding more tools.

The app already has many pages and tools. The next major product goal should be:

> Turn the EOAT Command Center from a collection of powerful pages into a guided engineering workflow system.

This means:

- Less tab-hopping
- More machine-centered context
- More guided audit flow
- Better completion logic
- Stronger validation
- Better evidence tracking
- Better PM tracking
- Better compatibility visibility
- Better release safety
- Faster loading
- Safer data handling

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3. Highest-Priority Feature Roadmap

Priority 1: Machine 360 Page

Create a new app-native page that gives a complete overview of one selected Press/Machine number.

This should become the main “single source of context” page for machine-level EOAT review.

Why This Matters

Right now, information is spread across:

- EOAT Audit
- Press View
- Notes
- Tags
- Open Items
- Photos
- Workbook Health
- KPI Dashboard
- FMEA
- Reports
- Robot Info workbook
- Compatibility rows

Machine 360 should combine all of this into one page.

Page Name

Recommended:

```
Machine 360
```

Alternative:

```
Press / EOAT 360
```

Add To Navigation

Add a new page spec:

```
key: machine_360
label: Machine 360
section: Capture or Analysis
factory_path: app.pages.machine_360:Machine360Page
```

Core Data Aggregation

Create a core service function that gathers all context for a selected machine:

```
core/project_data_service.py
```

or:

```
core/machine_360.py
```

Suggested API:

```python
def build_machine_360_context(project_root: str | Path, machine_number: str) ->
Machine360Context:
    ...
```

Data Sources

Machine 360 should gather from:

- EOAT Inventory physical audit rows
- Compatibility-derived rows
- Robot_Info.xlsx
- Annotation database
- Notes
- Tags
- Open items
- Photo index / photo folder links
- Validation findings
- KPI data if available
- FMEA data if available
- Reports that reference the machine if available

Required Cards / Sections

The page should show these cards.

Machine Identity

Show:

- Press/Machine #
- Plant/Area
- Press brand/model/tonnage if available
- Robot type
- Robot model/controller
- Tool #
- Part family

- Part name/description
- Cleanroom/non-cleanroom
- Entry type status

Physical Audit Summary

Show:

- Physical audit ID
- Audit date
- Auditor
- Status
- Priority
- Completion %
- Missing required fields
- Missing important fields
- Open Audit button

Compatibility Summary

Show:

- Compatible rows linked to this machine
- Source audit ID
- Compatibility source
- Compatibility source machine
- Fields propagated
- Compatibility conflicts
- Missing compatibility metadata
- Open Press View button

EOAT Tooling Summary

Show:

- EOAT Type
- EOAT Moves
- Connection Type
- Number of Parts Picked
- # of Vacuum Cups

- Cup Type/Material
- Cup Diameter/Size

• of Grippers

- Gripper Type
- Gripper Model
- Gripper Size
- Vacuum Generator Type
- Estimated EOAT Weight

Pneumatic Circuits

Show:

- EOAT Vacuum Circuits
- EOAT Pressure Circuits
- EOAT Interchangeable Circuits
- Robot Vacuum Circuits
- Robot Pressure Circuits
- Robot Interchangeable Circuits

Robot-side fields should come from Robot_Info.xlsx when available.

Sensors and Detection

Show:

- Sensors Present?
- Sensor Type
- Sensor Brand/Model
- Vacuum Confirmation Present?
- Part-Present Detection Present?
- Electrical/Wiring Present?
- Electrical Quick Disconnect Type
- Cable Management Condition

Mechanical / Routing

Show:

- Quick Disconnects Present?
- Pneumatic Quick Disconnect Type
- Tubing Condition
- Tubing Routing Notes
- Mounting Hardware Condition
- EOAT Alignment Condition
- Fastener/Locking Hardware Present?

Reliability / Performance

Show:

- Known Issues
- Drop/Mis-Pick History
- Maintenance Frequency
- Cycle Time Concern?
- Scrap/Quality Concern?
- Changeover Difficulty
- Pilot Candidate?

Documentation / Photos

Show:

- Drawing/CAD Available?
- BOM Available?
- Process Binder Complete?
- Spare Parts Identified?
- Photos Taken?
- Photo Folder/Link
- Required photo shot checklist status if available

Open Items

Show:

- Notes
- Tags
- Needs Review tags
- Info tags
- Follow-up actions
- Validation findings
- Open action items

Risk / FMEA

If available, show:

- Related failure modes
- Severity
- Frequency
- Detectability
- RPN
- Mitigation plan
- Open risks

KPI Signals

If available, show:

- Downtime
- Scrap
- Cycle time
- Part drop/mis-pick indicators
- Maintenance frequency
- Before/after pilot metrics

Action Buttons

Add:

- Open Audit
- Open Press View
- Add Note
- Add Tag
- Create Follow-Up
- Run Machine Validation
- Generate Machine Summary
- Open Photo Folder
- Generate PM Checklist
- Generate Work Instruction Draft

Performance Rules

Machine 360 must not deep-scan every file on every machine selection.

Use this approach:

1. Fast selection lookup first.
2. Use cached/indexed workbook data where available.
3. Only run expensive validation when the user clicks a button.
4. Debounce machine selector changes.
5. Show “last refreshed” and “data source” messages.

Acceptance Criteria

- Machine 360 appears in navigation.
- User can select or type a machine number.
- Page loads without requiring real private data.
- Works on demo project data.
- Shows physical audit rows separately from compatibility rows.
- Does not count compatibility rows as physical audits.
- Opens the correct audit when “Open Audit” is clicked.
- Opens Press View when “Open Press View” is clicked.

- Does not write files unless the user clicks a writing action.
- Has unit tests for context aggregation.
- Has a basic UI smoke test.

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4. Priority 2: Guided Audit Wizard

Goal

Create a step-by-step guided audit mode that helps the user complete audits faster and more truthfully.

The current full audit page should remain. The wizard should sit alongside it, not replace it.

Why This Matters

The existing audit form is powerful but dense. Guided Audit Mode should make data collection easier in the field.

The wizard should use existing field rules and Audit Coach logic instead of duplicating validation.

Recommended Location

Add either:

```
EOAT Audit > Guided Audit tab
```

or:

```
EOAT Audit > Start Guided Audit button
```

Wizard Steps

Step 1: Machine Identity

Fields:

- Audit ID
- Audit Date
- Auditor
- Plant/Area
- Press/Machine #
- Status
- Priority

- Follow-Up Needed

Actions:

- Generate Audit ID
- Lookup Machine
- Load Existing Audit
- Continue

Step 2: Robot / Tool Context

Fields:

- Robot Type
- Robot Model/Controller
- Tool #
- Part Family
- Part Name/Description
- Cleanroom/Non-Cleanroom

Actions:

- Apply machine lookup results
- Confirm robot info
- Continue

Step 3: EOAT Type and Tooling

Fields:

- EOAT Type
- EOAT Moves
- Connection Type
- Number of Parts Picked
- # of Vacuum Cups
- Cup Type/Material
- Cup Diameter/Size
- Vacuum Generator Type
- # of Grippers
- Gripper Type
- Gripper Model
- Gripper Size
- Estimated EOAT Weight

Visibility must follow `field_applies`.

Hybrid should show both vacuum and gripper details.

Miscellaneous should allow broad tooling fields where appropriate.

Step 4: Pneumatic Circuits

Fields:

- EOAT Vacuum Circuits
- EOAT Pressure Circuits
- EOAT Interchangeable Circuits
- Robot Vacuum Circuits
- Robot Pressure Circuits
- Robot Interchangeable Circuits

Rules:

- EOAT-side fields save to EOAT Master Tracker.
- Robot-side fields save to Robot_Info.xlsx.
- Robot fields should not cause extra writes if values are unchanged or blank/default-only.

Step 5: Sensors and Detection

Fields:

- Sensors Present?
- Sensor Type
- Sensor Brand/Model
- Vacuum Confirmation Present?
- Part-Present Detection Present?
- Electrical/Wiring Present?

Rules:

- If Sensors Present? is No, hide sensor detail fields.
- If Part-Present Detection Present? is Yes, default Sensor Type to Reed Switch and Sensor Brand/Model to SMC if allowed.
- If Electrical/Wiring Present? is No, hide wiring/cable fields.

Step 6: Connections / Routing / Mechanical

Fields:

- Quick Disconnects Present?
- Pneumatic Quick Disconnect Type
- Electrical Quick Disconnect Type

- Tubing Condition
- Tubing Routing Notes
- Cable Management Condition
- Mounting Hardware Condition
- EOAT Alignment Condition
- Fastener/Locking Hardware Present?

Rules:

- Quick disconnect details hide when Quick Disconnects Present? is No.
- Electrical Quick Disconnect Type hides if electrical/wiring is No.
- Tubing Routing Notes should be optional and should not reduce completion percentage if intentionally blank.

Step 7: Reliability / Performance

Fields:

- Known Issues
- Drop/Mis-Pick History
- Maintenance Frequency
- Cycle Time Concern?
- Scrap/Quality Concern?
- Changeover Difficulty

Actions:

- Create follow-up from known issue
- Tag issue
- Add note
- Link to FMEA candidate

Step 8: Documentation / Photos

Fields:

- Spare Parts Identified?
- Drawing/CAD Available?
- BOM Available?
- Process Binder Complete?
- Photos Taken?
- Photo Folder/Link

Rules:

- Documentation/photo defaults should default to No where already configured.
- If Photos Taken? is Yes and Photo Folder/Link is blank, warn.
- If Photos Taken? is No, show reminder but do not block save.

Step 9: Pilot / Final Notes

Fields:

- Pilot Candidate?
- Notes

Rules:

- Final Notes should be optional and should not reduce completion percentage if blank.

Step 10: Final Review

Show:

- Audit completion percentage
- Missing required fields
- Missing important fields
- Unknown / Not Checked fields
- Stale hidden conflicts
- Hybrid completeness warnings
- Semantic consistency warnings
- Photo evidence warnings
- Follow-up items that will be created
- Compatibility update impact if editing an existing audit
- Robot_Info update status preview

Actions:

- Save Audit
- Save Draft
- Mark Audit Complete Anyway
- Return to Step
- Open Full Audit

Manual Completion Feature

Add a manual completion flow:

```
Mark Audit Complete Anyway
```

This should not fake field values.

It should record:

- audit ID
- timestamp

- reason
- user-entered explanation
- remaining missing fields
- remaining unknown fields
- completion override status

Recommended storage:

- In workbook column if schema supports it:
- Manual Completion Override?
- Manual Completion Reason
- Manual Completion Timestamp
- Or in annotations/action log if workbook schema migration is not desired immediately.

Acceptance Criteria

- Full Audit form still works.
- Guided Audit Mode uses existing field rules.
- Guided Audit Mode uses existing Audit Coach summary.
- Hidden fields are not treated as missing.
- Unknown / Not Checked remains explicit.
- User can save draft from wizard.
- User can resume draft.
- User can save audit from wizard.
- Manual completion records reason and does not fabricate field values.
- Tests cover Vacuum, Mechanical / Gripper, Hybrid, Miscellaneous, and Unknown EOAT types.

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5. Priority 3: Workbook Truth Engine

Goal

Upgrade workbook validation from “partial validator” to a full truth engine.

The app depends heavily on workbook correctness. The validator should become a first-class safety and trust layer.

Recommended Name

```
Workbook Truth Engine
```

or:

```
Workbook Validation Engine
```

Core Responsibilities

Validate:

- Required sheets
- Required headers
- Unexpected duplicate headers
- Schema version
- Missing required fields
- Missing important fields
- Invalid dropdown values
- Duplicate Audit IDs
- Blank Audit IDs
- Duplicate physical rows
- Compatibility rows without source audit ID
- Compatibility rows without compatibility source
- Physical audit rows incorrectly marked compatible
- Compatibility rows counted as physical audits
- Broken photo folder links
- Photos marked Yes without links
- Photo links present while Photos Taken? is No
- Sensor detail fields filled when Sensors Present? is No
- Electrical fields filled when Electrical/Wiring Present? is No
- Quick disconnect details filled when Quick Disconnects Present? is No
- Vacuum fields filled on Mechanical / Gripper rows
- Gripper fields filled on Vacuum rows
- Hybrid rows missing both sides of tooling
- Robot-side circuit mismatches between audit and Robot_Info.xlsx
- Orphan notes
- Orphan tags
- Orphan open items
- Missing process binder documentation
- Missing CAD/BOM status
- Invalid priority/status values
- Old schema fields still present
- Legacy rows needing migration

Output Types

Generate:

1. Human-readable Markdown report.
2. JSON machine-readable validation report.
3. UI summary for Workbook Health.

4. Repair suggestion list.
5. Optional safe repair preview.

Repair Strategy

Do not auto-repair everything.

Classify repairs:

```
Safe automatic repair:
- clear stale hidden field to N/A
- normalize obvious dropdown casing
- add missing optional headers
- create missing report folders

Requires confirmation:
- migrate schema
- rename columns
- rewrite compatibility metadata
- repair duplicate Audit IDs
- modify robot info rows
- clear user-entered values

Never automatic:
- delete audit rows
- delete photos
- delete notes/tags
- delete reports
```

UI Requirements

Workbook Health should show:

- Overall status
- Critical blockers
- Warnings
- Info findings
- Findings grouped by machine
- Findings grouped by audit ID
- Findings grouped by field
- Filter by severity
- Open affected audit
- Open affected machine
- Export report
- Preview safe repairs
- Apply selected repair plan

Acceptance Criteria

- Workbook validator implementation status becomes implemented in tool registry.
- Existing Workbook Health page uses the new engine.
- Validator can run read-only.
- Validator can write reports.
- Validator never modifies workbook without confirmation.
- Tests cover stale hidden values, compatibility row issues, duplicate Audit IDs, broken photo links, and EOAT type semantic conflicts.

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6. Priority 4: GitHub Actions CI and Release Safety

Goal

Add automated checks so future changes do not break core behavior.

This repo is now large enough that local-only testing is risky.

Add Workflow

Create:

```
.github/workflows/ci.yml
```

Trigger

Run on:

```
push
pull_request
workflow_dispatch
```

Jobs

Test Job

Steps:

1. Check out repo.
2. Set up Python 3.11 or 3.12.
3. Install requirements.
4. Run pytest.

5. Run repo safety audit.
6. Run dashboard smoke test.

Example commands:

```
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python -m pytest
python scripts/repo_safety_audit.py --root .
$env:EOAT_COMMAND_CENTER_SMOKE_TEST="1"
python run_dashboard.py
```

For Linux, use:

```
QT_QPA_PLATFORM=offscreen
```

Additional CI Checks

Add tests for:

- Import every PageSpec factory path.
- Load tool registry.
- Ensure every implemented registry tool has a real module or documented reason.
- Ensure demo project mode does not require private files.
- Ensure config example loads.
- Ensure no real project root is committed.
- Ensure no generated reports are committed.
- Ensure no `.xlsx` operational workbooks are committed except explicitly allowed sanitized demo artifacts.

Acceptance Criteria

- CI passes on clean checkout.
- CI does not need private Nolato data.
- CI can smoke-launch the dashboard.
- CI runs repo safety audit.
- README documents CI and local test commands.

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7. Priority 5: PM Due Engine

Goal

Turn PM checklist generation into an actionable maintenance tracking system.

The existing PM checklist concept includes recurring inspection items like vacuum cups, pneumatic tubing, sensors, mounting hardware, EOAT alignment, quick disconnect fittings, and cable management.

The PM Due Engine should track what is due, overdue, completed, blocked, or upcoming.

Core Module

Create:

```
core/pm_due.py
```

Data Model

Define:

```python
@dataclass
class PMItem:
    id: str
    name: str
    frequency: str
    applies_to_eoat_types: list[str]
    category: str
    description: str
    evidence_required: bool
```

Define:

```python
@dataclass
class PMRecord:
    item_id: str
    machine_number: str
    audit_id: str
    last_completed_date: date | None
    next_due_date: date | None
    status: str
    notes: str
    completed_by: str
```

Default PM Items

Include:

- Inspect vacuum cups for wear/damage

- Inspect pneumatic tubing
- Verify sensor operation
- Inspect mounting hardware
- Verify EOAT alignment
- Check quick disconnect fittings
- Verify cable management condition
- Check gripper jaw/finger wear
- Check cylinder movement if cylinder fields exist
- Check robot-side pneumatic circuit labeling
- Check EOAT-side pneumatic circuit labeling
- Confirm process binder documentation

Status Values

Use:

```
not_started
due_soon
due
overdue
complete
blocked
not_applicable
```

UI

Extend PM Checklists page or create a new PM Due page.

Show:

- Due this week
- Overdue
- Completed recently
- Blocked
- Machine filter
- EOAT type filter
- Generate checklist
- Mark item complete
- Add notes
- Add photo evidence link
- Export PM pack

Storage

Store PM status in active project root outside repo.

Possible file:

```
00_Project_Admin/pm_due/pm_records.json
```

or workbook if preferred:

```
PM_Tracking.xlsx
```

JSON is easier for app use. Excel export can be generated.

Acceptance Criteria

- PM due status works with demo data.
- PM item applicability respects EOAT type.
- Vacuum-only PM items do not apply to pure gripper EOATs unless broad/misc type.
- Gripper-only PM items do not apply to pure vacuum EOATs.
- Hybrid EOATs include both vacuum and gripper PM items.
- PM completion writes to project root only.
- PM checklist export still works.

---

8. Secondary Feature: Photo Evidence Board

Goal

Make photo evidence structured instead of just storing random images.

Required Photo Types

Define required/optional shot types:

- Overall EOAT
- Vacuum Cups
- Grippers
- Tubing Routing
- Sensors
- Quick Disconnects
- Mounting Hardware
- Cable Management
- Wear / Damage
- Robot-Side Pneumatics
- EOAT-Side Pneumatics
- Tool Label / Identification

- Process Binder / Documentation Reference

Photo Evidence Status

For each audit, show:

```
Overall EOAT: present
Vacuum Cups: missing
Sensors: present
Quick Disconnects: present
Cable Management: missing
```

Rules

- If Photos Taken? is Yes but no required photo types are indexed, warn.
- If photo folder exists but Photo Index has no entries, warn.
- If photo files are missing on disk, warn.
- If photo type does not apply to EOAT type, mark not applicable.
- Hybrid should require both vacuum and gripper evidence where possible.

UI

Add to Photos page:

- Evidence checklist by audit
- Missing shot types
- Drag/drop photo import by shot type
- Rename/index photo
- Open folder
- Open photo
- Link photo to audit field

Acceptance Criteria

- Photo evidence board works with synthetic demo photos.
- Missing photo types appear in Audit Coach or Machine 360.
- Photo evidence status does not require private files in repo.
- Broken links are reported by Workbook Truth Engine.

---

9. Secondary Feature: Standardization Opportunity Finder

Goal

Find inconsistent components, naming, and spare part standardization opportunities.

Inputs

Use audit rows and BOM/spares data.

Analyze:

- Cup types
- Cup sizes
- Vacuum generators
- Gripper models
- Gripper types
- Sensor types
- Sensor brands/models
- Quick disconnect types
- Tubing condition patterns
- Connection types
- Spare parts identified status
- Missing BOMs
- Missing CAD
- Missing process binder documentation

Findings

Examples:

```
Finding:
The same cup material appears under 5 different names:
- Silicone
- Silicon
- Silcone
- Sili cup
- Rubber/Silicone

Recommendation:
Normalize to Silicone.
```

```
Finding:
7 EOATs use unknown sensor models.

Recommendation:
Create follow-up actions to identify sensor part numbers.
```

Output

Generate:

- Standardization opportunities report
- Component frequency table
- Unknown/missing part number table
- Suggested controlled vocabulary
- Recommended standard parts list
- Candidate BOM cleanup actions

UI

Add to BOM / Spare Parts page:

- "Find Standardization Opportunities"
- Top repeated components
- Unknown components
- One-off components
- Naming inconsistencies
- Export report

Acceptance Criteria

- Does not overwrite workbook.
- Produces Markdown and JSON report.
- Can run on demo data.
- Handles blank and unknown values safely.

---

10. Secondary Feature: Compatibility Matrix 2.0

Goal

Provide a matrix view of machine/tool compatibility.

Matrix Concept

Rows:

```
Machines / Presses
```

Columns:

```
Tools / Part Families / EOAT Source Audits
```

Cells:

```
Compatible
Not Compatible
Unknown
Conflict
Needs Review
```

Cell Details

Clicking a cell should show:

- Machine
- Tool/source audit
- Compatibility status
- Source audit ID
- Compatibility source
- Fields copied
- Conflicts
- Missing data preventing decision
- Recommended action

Filters

Add filters for:

- Plant/Area
- EOAT Type
- Robot Type
- Connection Type
- Part Family
- Vacuum cup count
- Gripper model
- Pneumatic circuit counts
- Quick disconnect type
- Status
- Priority

Acceptance Criteria

- Matrix separates physical audits from compatibility rows.
- Matrix does not create compatibility rows automatically.
- Matrix can export CSV/Markdown.
- Matrix can open relevant audit or Machine 360 page.

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11. Secondary Feature: Risk Heat Map

Goal

Turn FMEA-lite and issue data into a visual risk dashboard.

Failure Modes

Track:

- Vacuum loss
- Misalignment
- Tubing failure
- Sensor failure
- Mechanical wear
- Cable management issue
- Quick disconnect issue
- Drop/mis-pick
- Changeover error
- Documentation gap
- Pneumatic circuit mismatch
- Gripper wear
- Cylinder issue if cylinder fields are added

Heat Map

Rows:

```
Machines / EOATs
```

Columns:

```
Failure modes
```

Cell value:

```
Risk score or RPN
```

Click Behavior

Clicking a cell should show:

- Related audit fields
- Related notes
- Related tags
- Related photos
- Related open items
- FMEA entries
- Suggested mitigation
- Create action item button

Acceptance Criteria

- Uses existing FMEA and issue analysis data.
- Does not fabricate risk.
- Missing evidence should show as missing evidence, not as a fake score.
- Exports report.

---

12. Secondary Feature: Bad Actor Detector

Goal

Rank machines or EOATs that appear to cause the most issues.

Inputs

Use:

- Known Issues
- Drop/Mis-Pick History
- Priority
- Follow-Up Needed
- Tags
- Notes
- Open items
- Maintenance Frequency
- Scrap/Quality Concern?
- Cycle Time Concern?
- FMEA risk
- KPI data if available

Example Score

```
Bad Actor Score =
  issue_count * 2
+ high_priority_count * 3
+ critical_priority_count * 5
+ follow_up_count * 2
+ drop_history_weight
+ scrap_concern_weight
+ cycle_time_concern_weight
+ maintenance_frequency_weight
+ documentation_gap_penalty
```

Output

Show:

- Top 10 machines
- Top 10 EOATs
- Evidence for score
- Recommended next action
- Candidate pilot flag

Acceptance Criteria

- Score formula is visible to user.
- User can adjust weights later.
- Tool does not claim certainty if evidence is weak.
- Can export report.

---

13. Secondary Feature: Pilot ROI Estimator

Goal

Help select and justify pilot optimization projects.

Inputs

- Candidate machine
- Known issue severity
- Estimated downtime reduction
- Estimated scrap reduction
- Estimated cycle time improvement
- Maintenance reduction

- Cost of parts
- Cost of labor
- Implementation difficulty
- Confidence level
- Evidence notes

Outputs

- Qualitative ROI
- Payback estimate if cost data available
- Before/after KPI placeholders
- Confidence score
- Pilot recommendation paragraph
- Final report content block

UI

Add to Pilot Candidates page:

- Select candidate
- Enter assumptions
- Show ROI estimate
- Export pilot justification

Acceptance Criteria

- Clearly labels estimates as estimates.
- Does not invent financial values.
- Supports qualitative mode when dollar values are unavailable.
- Stores assumptions and timestamp.

---

14. Secondary Feature: Work Instruction Builder

Goal

Generate operator and technician work instruction drafts from actual audit data.

Work Instruction Types

Generate:

- Operator EOAT inspection checklist
- Technician EOAT troubleshooting guide
- EOAT rebuild checklist
- What to do on part drop
- What to check after changeover

- PM checklist
- Sensor verification guide
- Vacuum/gripper troubleshooting guide

Inputs

Use:

- EOAT Type
- Tooling fields
- Sensor fields
- Pneumatic circuit fields
- Known issues
- PM checklist items
- Photos
- Documentation status
- FMEA risks
- Open items

Outputs

- Markdown draft
- DOCX draft if existing report infrastructure supports it
- Checklist export
- Training/handoff package entry

Acceptance Criteria

- Does not claim missing documentation exists.
- Clearly marks missing photos or missing BOM/CAD.
- Pulls only from current project root.
- Supports demo data.

---

15. Secondary Feature: EOAT Change Validation Tool

Goal

Guide validation after EOAT repair, improvement, or pilot change.

Checklist

Include:

- EOAT mounted securely

- Vacuum/gripper function verified
- Sensor operation verified
- Quick disconnects verified
- Tubing/cable routing verified
- Dry cycle completed
- First-part pickup checked
- Drop/mis-pick checked
- Cycle time captured
- Scrap/quality concerns checked
- Photos captured
- Signoff completed

UI

Add to Pilot Candidates, PM Checklists, or Machine 360.

Actions:

- Start validation
- Save validation record
- Attach photos
- Create follow-up
- Generate validation report

Acceptance Criteria

- Validation records are stored in project root.
- Validation does not overwrite audit data unless user explicitly updates audit.
- Validation can be included in final handoff.

---

16. Secondary Feature: Data Import Wizard

Goal

Safely import external operational data without corrupting app data.

Supported Data Types

- Press capacity workbook
- Downtime export
- Scrap export
- Maintenance event export
- Cycle-time baseline
- Machine master list
- Robot list

- PM records

Workflow

1. Select file.
2. Detect file type.
3. Preview first rows.
4. Map columns.
5. Validate values.
6. Show import summary.
7. Confirm import.
8. Store mapping template.
9. Log import.

Data Safety

- Never commit imported data.
- Never overwrite existing data without backup.
- Keep raw imported files outside repo.
- Store import logs in project root.
- Allow dry run.

Acceptance Criteria

- Supports CSV and XLSX where practical.
- Handles missing columns gracefully.
- Writes import report.
- Can run dry-run preview without changes.

---

17. Secondary Feature: QR Code EOAT Labels

Goal

Generate QR labels for machines, EOATs, or audit records.

QR Contents

Could encode:

- Machine number
- Audit ID
- Local app lookup key
- Optional approved internal link
- Photo folder reference if approved

Use Case

Scanning a QR code opens or identifies:

- Machine 360
- Audit record
- PM checklist
- Photo evidence page

Safety

Do not encode private operational details into QR unless approved.

Preferred QR value:

```
eoat://machine/123
```

or:

```
eoat://audit/EOAT-2026-0001
```

Then the app resolves the value locally.

Acceptance Criteria

- QR generation is optional.
- QR content is minimal.
- No private data is exposed unnecessarily.
- Labels can be exported as printable PDF or image sheet if dependencies support it.

---

18. Secondary Feature: “What Changed?” Timeline

Goal

Show audit and machine history over time.

Timeline Events

Track:

- Audit created
- Audit updated

- Field changed
- Compatibility rows updated
- Robot_Info updated
- Note added
- Tag added
- Follow-up created
- PM checklist generated
- Report generated
- Validation finding created
- Manual completion override applied
- Photo evidence added

UI

Available from:

- Machine 360
- Audit page
- Press View

Acceptance Criteria

- Timeline uses existing activity logs where possible.
- Does not expose debug-only stack traces.
- Can filter by event type.
- Can export timeline to Markdown.

---

19. Secondary Feature: App Health Doctor

Goal

Create a user-facing troubleshooting page that checks whether the app/project environment is healthy.

Checks

Include:

- Python version
- Dependency availability
- PySide import
- Project root exists
- Demo mode vs real mode
- Master workbook exists
- Workbook lock/open status
- Required folders exist

- Config file valid
- Robot_Info.xlsx exists or is optional
- Annotation database available
- Photo index available
- Scheduled tasks installed
- Scheduled task last result
- Scheduled tool log exists
- Performance log exists
- Dashboard cache status
- Repo safety audit status
- Git available
- Release readiness status

Output

Use status levels:

```
Green: OK
Yellow: Needs attention
Red: Blocking
Gray: Not checked
```

Acceptance Criteria

- Does not modify project files unless explicitly running a repair.
- Provides recommended action for each warning/error.
- Links to relevant app pages.

---

20. Architecture Refactoring Recommendations

20.1 Create Project Data Service

Create:

```
core/project_data_service.py
```

Purpose:

Provide a clean data-access interface for pages.

Suggested methods:

```python
class ProjectDataService:
    def __init__(self, project_root: str | Path): ...

    def get_audit(self, audit_id: str): ...
    def list_audits(self): ...
    def list_machines(self): ...
    def get_machine_context(self, machine_number: str): ...
    def get_machine_360(self, machine_number: str): ...
    def get_open_items_for_machine(self, machine_number: str): ...
    def get_photos_for_audit(self, audit_id: str): ...
    def get_validation_findings(self, scope: str, target_id: str): ...
```

Pages should avoid directly re-implementing workbook/annotation/report lookup logic.

20.2 Add SQLite Shadow Index

Excel should remain the official handoff output, but the app should use a local SQLite shadow index for faster reads.

Create:

```
project_data/eoat_index.sqlite
```

Potential tables:

- audits
- compatibility_entries
- machines
- robot_info
- photos
- issues
- notes
- tags
- open_items
- validation_findings
- kpi_snapshots
- schema_versions

Rules:

- Rebuild index from workbook when stale.
- Never treat index as more authoritative than workbook unless explicitly designed later.
- Store index in project root, not repo.
- Use index for fast reads and search.

20.3 Formal Schema Migration System

Create:

```
core/schema_migrations.py
```

Purpose:

Manage workbook schema changes safely.

Features:

- Detect current schema version.
- Preview migration.
- Backup workbook before migration.
- Apply migration.
- Validate after migration.
- Log migration.

Example migrations:

- Rename Number of Vacuum Cups to Number of Parts Picked.
- Add # of Vacuum Cups.
- Add # of Grippers.
- Add Gripper Type.
- Add Gripper Model.
- Add Electrical/Wiring Present?
- Add pneumatic circuit fields.
- Add manual completion fields.
- Add cylinder fields if needed.

20.4 Split Audit Page Further

Current audit page is powerful but large.

Recommended structure:

```
app/pages/audit/
  page.py
  field_builder.py
  field_visibility_controller.py
  draft_controller.py
  lookup_controller.py
  save_controller.py
  annotation_controller.py
```

```
  guided_wizard.py
  compatibility_tab.py
  interview_tab.py
```

Keep backward-compatible imports if needed.

20.5 Add Typed Navigation Actions

Create:

```
app/navigation_actions.py
```

Avoid scattered string actions.

Use typed action objects for:

- open audit
- open machine
- open note
- open tag
- open report
- open photo
- open validation finding

20.6 Standardize Page Refresh Contract

Create a standard page protocol.

Preferred method:

```python
def refresh_page(self, reason: RefreshReason) -> RefreshResult:
    ...
```

Avoid relying forever on a chain of optional method names like:

```
refresh_status
refresh_metrics
refresh
refresh_data
```

Keep fallback support temporarily.

20.7 Add Refresh Cost Levels

Define:

```
cheap
workbook_read
workbook_write
deep_analysis
external_process
```

Events should specify allowed refresh cost.

Example:

```
emit(EVENT_AUDIT_SAVED, {"refresh_mode": "invalidate_only",
"allowed_refresh_cost": "cheap"})
```

Pages should not do deep workbook scans after small events unless explicitly requested.

---

21. Settings Page Expansion Ideas

The Settings page should become more useful and less like a storage closet where dropdowns go to retire.

21.1 Replace Fixed Audit Defaults With Default Rules

Current audit defaults should become user-managed default rules.

Add:

```
New Default +
```

Workflow:

1. Click New Default.
2. Select field.
3. Select condition, optional.
4. Enter default value.
5. Save to defaults list.

Example Default Rules

```
Field: Vacuum Confirmation Present?
Condition: EOAT Type is Vacuum or Hybrid
Default: Yes
```

```
Field: Part-Present Detection Present?
Condition: always
Default: No
```

```
Field: Sensor Type
Condition: Part-Present Detection Present? is Yes
Default: Reed Switch
```

```
Field: Sensor Brand/Model
Condition: Part-Present Detection Present? is Yes
Default: SMC
```

UI

Show defaults as a table:

Enabled	Field	Condition	Default	Source	Actions

Actions:

- Edit
- Duplicate
- Disable
- Delete
- Move up/down
- Test against current form

Acceptance Criteria

- Existing defaults still load.
- New defaults are saved to local config.
- User can add/edit/delete defaults.
- Defaults are not applied over meaningful user-entered values.
- Defaults should not trigger false unsaved changes on a clean new form.

21.2 Scheduled Report Settings

Add settings for:

- Daily report enabled
- Daily report days
- Daily report time
- Weekly report enabled
- Weekly report day
- Weekly report time
- Timezone
- Dry-run mode
- Catch-up behavior
- Duplicate report behavior
- Open scheduled log
- Install/repair scheduled tasks
- Uninstall scheduled tasks
- Test scheduled daily report
- Test scheduled weekly report

21.3 Backup Settings

Add settings for:

- Backup enabled
- Backup before workbook write
- Backup before schema migration
- Backup retention count
- Backup retention days
- Backup folder
- Include config backup
- Include reports backup
- Include annotation database backup
- Preview cleanup
- Run cleanup

21.4 Data and Cache Settings

Add settings for:

- Dashboard cache enabled
- Rebuild cache
- Clear cache
- Rebuild SQLite shadow index
- Clear SQLite shadow index
- Open performance log
- Open cache folder

- Enable verbose performance logging
- Log rotation limit

21.5 Safety / Release Settings

Add settings for:

- Run repo safety audit
- Open repo safety report
- Check ignored files
- Check local config safety
- Check demo mode
- Check staged files
- Release readiness checklist

---

22. Data Completion Improvements

22.1 Accepted Unknowns

Add a way to distinguish:

- Unknown because not checked yet
- Unknown because unavailable
- Unknown but accepted
- Unknown requiring follow-up

Recommended approach:

Do not add too many dropdown values directly. Instead, store disposition metadata:

```
Field: Sensor Brand/Model
Value: Unknown / Not Checked
Disposition: Accepted Unknown
Reason: Label unreadable
Follow-up needed: No
Timestamp: ...
```

Audit Coach should treat accepted unknowns differently from unfinished unknowns.

22.2 Manual Completion Override

Add manual audit completion.

Rules:

- Do not fake blank fields.
- Record who/when/why.
- Keep missing fields visible in history.
- Exclude intentionally optional fields from completion percentage.
- Show override clearly in Workbook Health and Machine 360.

22.3 Optional Field Rules

Some fields should not reduce completion percentage when blank:

- Tubing Routing Notes
- Final Notes
- Any future purely informational comments
- Cylinder section if all cylinder cells are blank
- Other fields explicitly marked optional

Create a field metadata system:

```
FieldCompletionPolicy:
    required
    important
    optional
    optional_if_blank
    excluded_from_completion
    required_when_condition
```

---

23. Future Field Expansion: Cylinder Support

If not already implemented, support cylinder fields in EOAT Type and Tooling.

Fields:

- **of Cylinders**

- Cylinder Type

Cylinder Type dropdown:

- Linear
- Rotary

Default:

```
Linear
```

Rules:

- Cylinder section should always be visible.
- If both cylinder cells are empty, they should not count against Audit Coach completion.
- If one cylinder field is filled, the other should become important/missing.
- Cylinder fields should appear in Machine 360, PM Due Engine, FMEA, and Standardization Opportunity Finder.

---

24. Tag System Expansion

Add tag category:

```
Info
```

Tags should not only represent problems. They should also support informational labeling.

Tag categories should include:

- Problem
- Risk
- Follow-Up
- Info
- Documentation
- PM
- Pilot
- Safety
- Quality
- Compatibility

Info tags should not automatically imply action needed.

Example Info tags:

- Robot side shared circuit
- No electrical on EOAT
- Uses uncommon gripper
- Photos confirmed
- Process binder updated
- Confirmed with technician

25. UX Improvements Across App

25.1 Global Command Palette Expansion

The command palette should support:

- Open machine
- Open audit
- Create note
- Create follow-up
- Run validation
- Generate daily summary
- Generate PM checklist
- Open scheduled report log
- Rebuild cache
- Open settings default rules

25.2 Global Search Improvements

Search should return:

- Audits
- Machines
- Notes
- Tags
- Open items
- Reports
- Photos
- Validation findings
- PM records
- FMEA entries
- Pilot candidates

Each search result should have a typed navigation action.

25.3 User Guidance Text

Every major page should answer:

```
What is this page for?
What should I do here?
What files/data does this page use?
Does this page modify project files?
```

25.4 Evidence Confidence Labels

Reports and dashboards should show confidence:

```
Strong evidence
Moderate evidence
Weak evidence
Missing evidence
```

Avoid overstating conclusions when fields are blank or unknown.

25.5 “Next Best Action” System

Home and Machine 360 should recommend next actions:

Examples:

- Complete missing required audit fields for Press 12.
- Add photo folder link for EOAT-2026-0041.
- Review compatibility conflict on Machine 1.
- Run workbook validation after schema migration.
- Generate Friday weekly summary.
- Identify sensor model for high-priority follow-up.

---

26. Reporting Improvements

26.1 Machine Summary Report

Generate report for one machine:

- identity
- audit status
- tooling
- circuits
- sensors
- issues
- photos
- open items
- validation findings
- PM status
- recommendations

26.2 Weekly Engineering Brief

Current weekly summaries should be expanded to include:

- New audits
- Completed audits
- Machines with new warnings
- Top open items
- Scheduled report status
- Bad actor changes
- PM overdue count
- Pilot candidate changes
- Documentation gaps closed
- Decisions needed

26.3 Leadership Summary

Create a concise leadership report:

- Total audits completed
- Percent complete
- Top risks
- Top pilot candidates
- KPI baseline status
- PM standard progress
- Documentation gap progress
- Next week plan
- Decisions needed

26.4 Final Handoff Index

Final Handoff should generate an index that links:

- Final master tracker
- Robot Info workbook
- FMEA
- KPI dashboard
- PM checklist package
- BOM/spares report
- Standard design guidelines
- Work instructions
- Pilot report
- Training materials
- Photos/evidence
- Open issues
- Recommendations

27. Implementation Rules For Codex

Rule 1: Do Not Break Existing Pages

Existing pages must continue working:

- Home
- EOAT Audit
- Press View
- Notes
- Tags
- Open Items
- Photos
- Audit Progress
- Issue Analysis
- FMEA
- Pilot Candidates
- KPI Dashboard
- Standards Docs
- PM Checklists
- BOM / Spare Parts
- Reports
- Scheduled Reports
- Final Handoff
- Workbook Health
- Performance
- Backup Manager
- Release Readiness
- Settings

Rule 2: Keep Real Data Out Of Repo

No private operational data should be committed.

Do not commit:

- real workbooks
- real audit reports
- real photos
- capacity files
- downtime/scrap/cycle-time files
- mold numbers
- part numbers
- customer names
- internal paths

- local config
- generated cache/log files
- backups

Rule 3: Prefer Core Logic Over UI Logic

UI pages should call core functions.

Avoid putting workbook parsing, validation, scoring, and report generation directly in page files.

Rule 4: Write Tests

Each new feature should include core tests.

At minimum:

- data aggregation tests
- field applicability tests
- save/preview tests
- demo project tests
- no-private-data assumptions

Rule 5: Use Background Tasks For Slow Work

Any workbook-heavy, report-heavy, or scan-heavy operation should run through the background task system.

Rule 6: Avoid Surprise Writes

Any action that modifies files should clearly indicate that it modifies files.

Use confirmations for:

- schema migrations
- repair actions
- deleting/cleaning backups
- updating compatibility rows
- rewriting workbook values

Rule 7: Separate Preview From Apply

For repairs, migrations, imports, compatibility changes, and generated outputs:

1. Preview
2. Confirm

3. Apply
4. Report result

Rule 8: Preserve Truthfulness

Do not hide uncertainty.

If data is missing, say it is missing.

If a row is compatibility-derived, label it as compatibility-derived.

If a value is unknown, keep it unknown.

If evidence is weak, say evidence is weak.

Do not invent KPI improvements, ROI values, or maintenance conclusions.

---

28. Recommended Implementation Order

Phase 1: Safety and Infrastructure

1. Add GitHub Actions CI.
2. Pin or bound dependency versions.
3. Add PageSpec import test.
4. Add tool registry validation test.
5. Add repo safety audit to CI.
6. Add dashboard smoke test to CI.

Phase 2: Core Data Service

1. Create ProjectDataService.
2. Add machine list retrieval.
3. Add audit retrieval.
4. Add machine context aggregation.
5. Add tests using demo data.
6. Keep current pages working.

Phase 3: Machine 360

1. Add Machine 360 page spec and nav.
2. Build basic machine selector.
3. Show identity and audit cards.
4. Add compatibility card.
5. Add robot info card.

6. Add notes/tags/open items card.
7. Add photo/documentation card.
8. Add action buttons.
9. Add tests.

Phase 4: Workbook Truth Engine

1. Promote workbook validator to implemented.
2. Add structured validation findings.
3. Add grouped findings.
4. Add machine/audit scoped validation.
5. Add repair preview system.
6. Add UI support in Workbook Health.
7. Add tests.

Phase 5: Guided Audit Wizard

1. Add Guided Audit sub-tab.
2. Use existing field rules.
3. Use Audit Coach summaries.
4. Add step navigation.
5. Add final review.
6. Add manual completion override.
7. Add tests.

Phase 6: PM Due Engine

1. Add PM data model.
2. Add due date logic.
3. Add PM records storage.
4. Extend PM Checklists page.
5. Add Machine 360 PM card.
6. Add export.
7. Add tests.

Phase 7: Evidence and Standardization

1. Add Photo Evidence Board.
2. Add Standardization Opportunity Finder.
3. Add Compatibility Matrix 2.0.
4. Add Bad Actor Detector.
5. Add Risk Heat Map.

Phase 8: Handoff Enhancements

1. Add Work Instruction Builder.
2. Add Pilot ROI Estimator.

3. Add EOAT Change Validation Tool.
4. Add Final Handoff Index improvements.
5. Add Leadership Summary.

---

29. Small Kickoff Prompt For Codex

Use this short prompt to start the next coding session:

```
I want you to begin implementing the next major EOAT Command Center expansion
using the attached feature/implementation plan.

Start with Phase 1 and Phase 2 only:
1. Add GitHub Actions CI / release safety checks if missing.
2. Add a ProjectDataService core layer that can aggregate machine-level context
from the existing demo project/workbook structures.
3. Do not build all features yet.
4. Do not remove or break existing pages.
5. Do not commit real data or generated private files.
6. Add tests for the new service and any safety checks.
7. Keep the implementation local-first, deterministic, and compatible with the
current PySide dashboard architecture.

After Phase 1 and Phase 2 are working, stop and summarize what changed, what
tests pass, and what should be implemented next.
```

---

30. Final Product Vision

The final app should feel like this:

```
Home:
What needs attention today?

Machine 360:
What is true about this press/machine?

Guided Audit:
Help me capture clean audit data.

Workbook Health:
Can I trust the data?
```

```
PM Due:
What maintenance is due?

Pilot Candidates:
Where should improvement effort go?

Reports:
What do I need to tell people?

Final Handoff:
What evidence proves the project is complete?
```

That is the direction.

Not more random tools.

Not more scattered buttons.

A guided, truthful, local-first engineering command center for EOAT standardization and optimization.